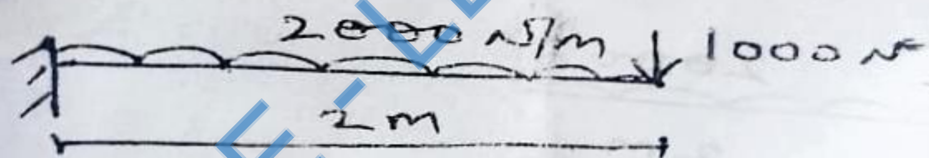
MID TERM II (B.E. III SEM 2022-23)
ME:204a Mechanics of Solids
MECH.ENGG. DEPT., MBM UNIVERSITY

TIME: 60 MIN MM:20

1.At a point within a body subjected to two mutually perpendicular stresses 80 N/mm^2 and 40 N/mm^2 tensile and 60N/mm^2 of shear stress. Find σ_t, σ_n and σ_r on a plane inclined at 45° to normal cross-section. Solve with Mohr's circle method.

2.Find deflection at the free end as in fig1. If the section is rectangular 100 mm x 200 mm and E=2X10^5 N/mm^2. (8)



3.Derive Torsion equation and state the assumptions made. (6)

OR

4.Determine ratio of breaking strength of two columns one hollow and other solid. Both are made of same material and have same length, area and end conditions. The internal dia of hollow column is half of external dia. (6)

MBM University Jodhpur

Department of mechanical engineering

Time: 1 Hr | Subject- MMI | Max Marks-10

Q.1 A strain gauge bounded to a steel beam of 0.20 meter long and has a cross sectional area of 0.4×10^{-3} meters. Young's modulus of elasticity for steel is 200 GN/m^2. The strain gauge has an unstrained resistance of 120 ohm and a gauge factor of 2. When load is applied the gauge resistance changes by 0.012 ohm. Calculate the change in length of steel beam and the amount of force applied to the beam. 2M

Q.2 Explain gauge factor of a strain gauge and obtain its expression. 3M

Q.3 Explain construction and working of the following transducers with neat sketch. 3M

a. Magnetic flow meter (2M) b. RTD and Thermistor

2^{nd} Midterm

ME 201A: Engineering Thermodynamics

Date 02/01/2022

Exam Duration: One Hour

Maximum Marks: 20

1. Write down the first and second TdS equations, and derive the expression for the difference in heat capacities, C_p and C_v. What does the expression signify? [7]
2. Derive the expression for volumetric efficiency of a single stage reciprocating compressor. Discuss how the volumetric efficiency varies with clearance ratio. [7]
3. Explain the analysis of flue gases by orsat apparatus with diagram. [6]

MBM University, Jodhpur

Department of Mechanical Engineering

B.E. IInd Year IIIrd Semester

Second Mid-Term Examination Session 2022-23

ME-203A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 20**

Q. 1. The four bar mechanism having dimensions as follows: $AB = 300$ mm, $BC = 360$ mm, $CD = 360$ mm, and $AD = 600$ mm. Here, AD is fixed with ground. Angle $\angle DAB$ is $60°$ and crank AB is rotates 300 rpm and angular acceleration of 20 rad/sec^2 both in clockwise direction. Determine the linear velocity, angular velocity and angular acceleration of BC and CD. (10)

Q. 2 (a) A rope dives transmits 800 kW from a pulley of effective diameter 3 m, which runs at a speed of 120 rpm. The angle of lap is 165°; the angle of groove 52°; the coefficient of friction 0.25; the mass of rope 2.5 kg/m and the allowable tension in each rope 2500 N. Find the number of ropes required to transmit 800 kW power. (6)

(b) Obtain an expression for the length of chain. (4)

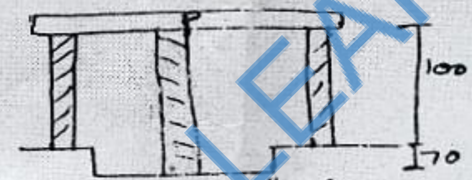
MID TERM 1 (B.E. III SEM 2022-23)

ME:204a Mechanics of Solids

MECH.ENGG. DEPT., MBM UNIVERSITY

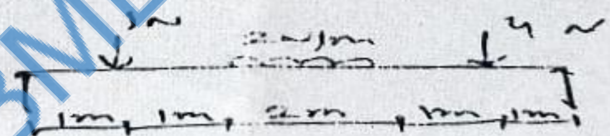
TIME: 60 MIN MM:20

1. Two brass rods and one steel rod together support a load as in fig.. If the stresses in bars and steel are not to exceed 60 Nlmm^2 and 120 Nlmm^2. find the safe load that can be supported. E Steel = 2 x 10^5 Nlmm^2 E bars = 1 x 10^5 Nlmm^2, Area Steel = 1500 mm^2, Area brass = 1000 mm^2



(6)

2. Draw the SFD and BMD for the following:



(8)

3. Derive bending equation and state the assumptions made.

OR

3. A timber joist of 6 meters span has to carry a load of 500 kg/m. Find the dimensions of joist, if the maximum permissible stress is limited to 80 Kg/cm^2. The depth of joist has to be twice the width.

"MATERIAL TECHNOLOGY" I MID-TERM EXAM 2022-23

Date: 21/11/2022

MM: 20

1. Sketch and explain all type of crystal structures. (5)
2. Define the term: - number of attoms in unit cell, coordination number and atomic packing factor with examples. (5)
3. Draw a neat sketch of $Fe_3C - C$ diagram labeling all important phases and constituents. (5)
4. Explain and sketch different dislocations. (5)

1st Midterm

ME 201A: Engineering Thermodynamics

Date 21/11/2022

Exam Duration: One Hour

Maximum Marks: 10

1. A body of constant heat capacity c_p and initial temperature T_1 is placed in contact with a heat reservoir at temperature T_2 and comes to thermal equilibrium with it. If $T_2 > T_1$, calculate the entropy change of the universe and show that it is always positive. [2]
2. Derive the expression of availability or maximum work done by the closed system in the change of state from 1 to 2 by interacting only with the surroundings at p_o, T_o. [5]
3. Derive the vander Waals constants in terms of critical properties. [3]

MBM University Jodhpur

Department of mechanical engineering

Time: 1 Hr | Subject- MMI | *Max Marks-10*

Q.1 Explain the importance of impedance loading and matching for a measuring instrument. 3M

Q.2 Explain the working principle of Piezo-electric accelerometer and Inductive pick-up tachometer. 4M

Q.3 Explain the following terms

a. Hysteresis and dead zone b. Precision and repeatability C. Sensitivity and Threshold 3M

MBM University, Jodhpur

Department of Mechanical Engineering

B.E. IInd Year IIIrd Semester

First Mid-Term Examination Session 2022-23

ME-203A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 20**

Q. 1. (a) Discuss the effect of slip and creep of belt on the pulleys on the velocity ratio of a belt drive. (3)

(b) Derive the relation for the ratio of belt tensions in a flat-belt drive. (3)

(c) A casting having a mass of 100 kg is suspended freely from a rope. The rope makes 2 turns round a drum of 300 mm diameter rotating at 24 rpm. The other end of the rope is pulled by a man. Calculate the force required by a man, power to raise the casting. Take $\mu = 0.3$. (4)

Q. 2. (a) Sketch and explain the various inversion of four-bar chain mechanism. (4)

(b) Explain different kinds of kinematic pairs giving example for each one of them. (6)